Reimagining Catalogs and Repositories to Enhance the FAIRness of Research Data

Toward a Shared Catalog for Integrated Data Discovery

John E Marcotte, PhD, Sarah E Rush, MSW, Kelly Ogden-Schuette, MIA

Who does this impact?

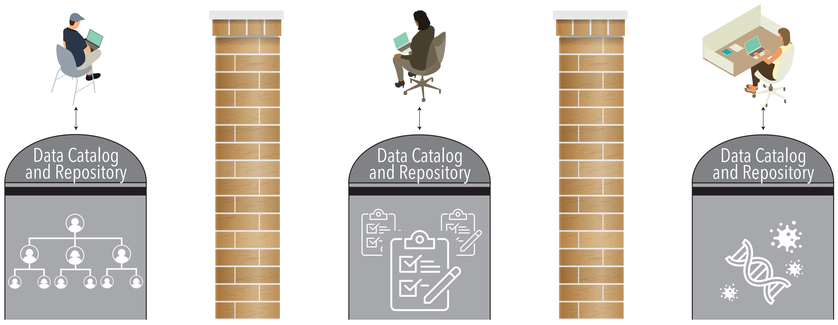
- **Librarians & Archivists** – Support users in finding data more efficiently
- **Researchers** – Easier access to diverse and distributed datasets
- **IT and Archive Professionals** – Build systems that talk to each other
- **Funders** – Support initiatives that improve scientific efficiency and innovation

The Problem

- Data associated with a single study may be fragmented across multiple repositories
- Metadata standards vary, making cross-repository discovery difficult
- Repositories are often domain-specific, which hampers interdisciplinary research
- High value data remain underutilized because discoverability tools are insufficient

Why This Matters

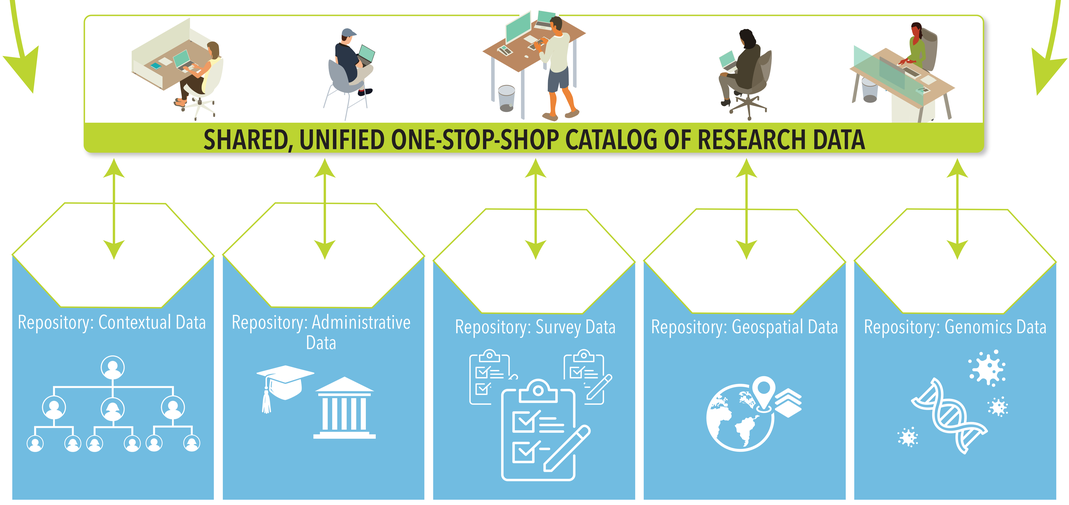
- Supports F.A.I.R. (Findable, Accessible, Interoperable, Reusable) best practices for data
- Increases data discoverability and interoperability across disparate repositories that often exist to support specific types of data
- Discoverable data are findable in a variety of catalogs, support study comparisons and have enhanced metadata to facilitate searchers
- Boosts data reuse and transparency
- Helps researchers find complementary datasets – even across domains
- Makes it easier to compare study content (study or variable metadata) across studies, even when data are in separate locations



The Vision: Bridging Silos and Gaps

Imagine a "One Stop Shop" for metadata: a shared, centralized catalog that surfaces key details of studies, regardless of where the actual data live.

Researchers would continue to deposit in domain-appropriate repositories and a centralized catalog would enable cross-study and cross-repository comparisons. For example, GxE (genomics x environment) research would be enhanced by bridging survey and genomic data.



Technical & Policy Considerations

- Metadata harmonization via APIs (e.g., ICPSR's Object Export API)
- Handle mirror deposits (same data in multiple places)
- Keep the catalog current via automated updates (e.g., scheduled syncs)
- Address access policies and permissions (not all data are open)

Takeaways

- **Improves Interoperability:** A shared catalog bridges gaps between siloed repositories
- **Increases Reusability:** Researchers get more value from existing data
- **Amplifies Findability:** Librarians and archivists gain new tools to support data discovery

Use Cases

Example 1: The Future of Families and Child Wellbeing Study (FFCWS)
Public data: DSDR & Princeton
Restricted data: Princeton
Genomic data: dbGaP
Uses MeSH for metadata tagging

Example 2: Los Angeles Family and Neighborhood Survey (L.A.FANS)
Survey and Biomarker data: DSDR
Contextual data: Census / ACS

M | ICPSR INSTITUTE FOR SOCIAL RESEARCH UNIVERSITY OF MICHIGAN

DSDR A Population Health Data Archive Data Sharing for Demographic Research

NACDA National Archive of Computerized Data on Aging

Learn More!
https://myumi.ch/ICPSR-DSDR